Meaningfully reducing consumption of meat and animal products is an unsolved problem: A meta-analysis

Network for Sustainable Food Transitions (NEST)

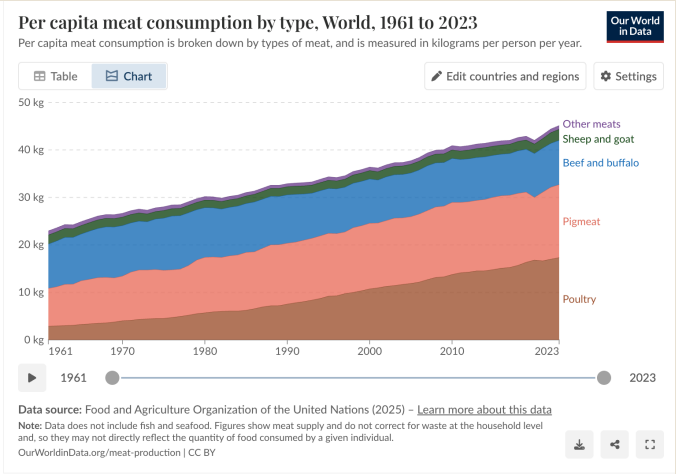
Seth Green

Humane & Sustainable Food Lab, Stanford University

2026-03-26

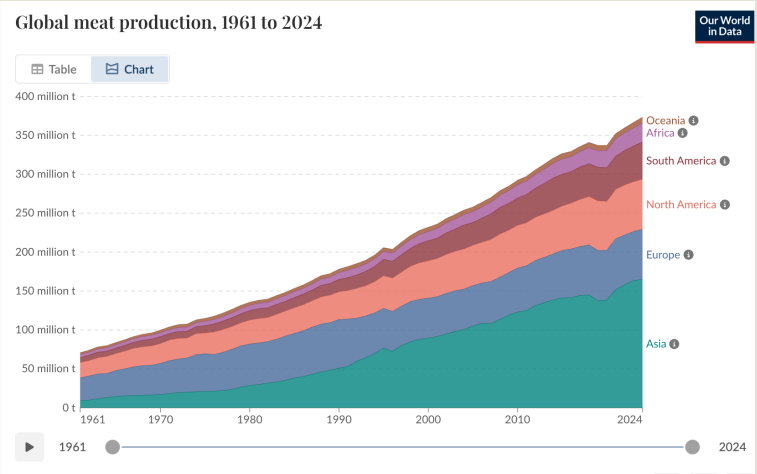
HUMANE & SUSTAINABLE FOOD LAB
AT STANFORD UNIVERSITY

Fact 1: meat consumption is rising across every category



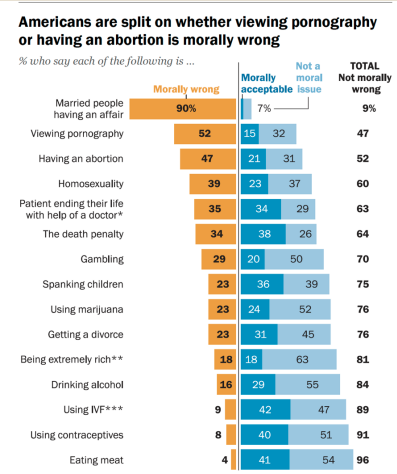
HUMANE & SUSTAINABLE FOOD LAB
AT STANFORD UNIVERSITY

. . . everywhere



HUMANE & SUSTAINABLE FOOD LAB
AT STANFORD UNIVERSITY

Fact 2: people think eating meat is fine



HUMANE & SUSTAINABLE FOOD LAB
AT STANFORD UNIVERSITY

Fact 3: transformational effects don't get discovered in the social sciences because they're already obvious

Daniel Lakens, talking about a putative effect size of d = 2 (the effect of hunger on judge's behavior):

> *"If hunger had an effect on our mental resources of this magnitude, our society would fall into minor chaos every day at 11:45... Just like manufacturers take size differences between men and women into account when producing items such as golf clubs or watches, we would stop teaching in the time before lunch, doctors would not schedule surgery, and driving before lunch would be illegal.* ***If a psychological effect is this big, we don't need to discover it and publish it in a scientific journal — you would already know it exists.****"*

HUMANE & SUSTAINABLE FOOD LAB
AT STANFORD UNIVERSITY

So how about this meta-analysis

- I worked on a meta-analytic review of interventions to get people to eat less MAP.
- We only looked at rigorous studies (RCTs with > 50 subjects that measured MAP consumption directly at least a day after treatment began)
- As of 12/2023, 35 papers met our criteria

Full Length Article

Meaningfully reducing consumption of meat and animal products is an unsolved problem: A meta-analysis

Seth Ariel Green [a], Benny Smith [b], Maya B. Mathur [a]

Show more

+ Add to Mendeley Share Cite

https://doi.org/10.1016/j.appet.2025.108233

Get rights and content

Under a Creative Commons license

Open access

HUMANE & SUSTAINABLE FOOD LAB
AT STANFORD UNIVERSITY

What interventions have been tried?

- Almost everything tries to:
  - Tell people meat is bad (environmental/health/animal welfare)
  - Make plant-based options more salient (eye-level on menus)
  - Make animal-based options less convenient
  - Tell people plant-based options are tasty and popular
  - Exploit psychological tendencies (social norms, commitment)
- Overall, small & sparse effects

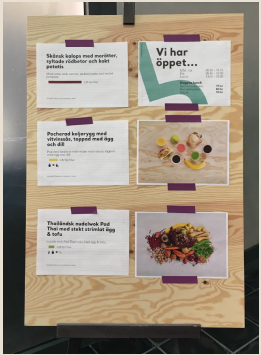
Table 1. Main findings.

Approach	N (Studies)	N (Estimates)	SMD	τ	95% CIs	p val
Overall	41	112	0.07	0.08	[0.02, 0.12]	0.007
Theory						
Choice Architecture	2	3	0.21	0	[-0.99, 1.42]	0.267
Psychology	9	12	0.11	0.03	[-0.03, 0.24]	0.092
Persuasion	25	77	0.07	0.09	[0.01, 0.13]	0.023
Persuasion & Psychology	11	20	0.11	0.19	[-0.06, 0.28]	0.189
Types of Persuasion						
Animal Welfare	16	65	0.03	0.04	[-0.02, 0.07]	0.189
Environment	16	29	0.09	0.12	[-0.02, 0.19]	0.095
Health	18	29	0.08	0.09	[-0.01, 0.17]	0.064

HUMANE & SUSTAINABLE FOOD LAB
AT STANFORD UNIVERSITY

Choice architecture in particular: very few credible designs

- Only 2 studies, 3 estimates in our sample
- Most studies lack real-world follow-up: we don't know if cafeteria-based nudges change what people actually buy over time
- Lab and online studies dominate; field evidence is thin
- Pooled SMD = 0.21, but CI includes 0 ($[-0.99, 1.42]$) because there are, again, two papers



example menu from Andersson, O., & Nelander, L. (2021). Nudge the lunch: A field experiment testing menu-primacy effects on lunch choices. *Games*, 12(1), 2.

HUMANE & SUSTAINABLE FOOD LAB
AT STANFORD UNIVERSITY

Hi Isabel :)

- Time to stir the pot!
- What you call field experiments, I would call field *studies*. An experiment needs random assignment IMHO
- Looking through your citations, *are* there any experiments that measure outcomes with any delay at all? I think Ginn & Sparkman (2024) is the only one that even theoretically would qualify and I think it's outside your time period
- was also surprised not to see Coker et al. (2022), "A dynamic social norm messaging intervention to reduce meat consumption: A randomized cross-over trial in retail store restaurants"

HUMANE & SUSTAINABLE FOOD LAB
AT STANFORD UNIVERSITY

but most importantly

- It's really hard to gauge how much we should buy the causal assumptions in non-randomized before-after papers
- Often it's clear that a lot of things changed, e.g. the number of meals served changes dramatically between periods
- Question for Isabel/anyone: *should* we grant these identification strategies the benefit of the doubt? I say no, the default is to assume bias, and effect sizes are typically small so a little bias can easily create type S (sign) errors.
- What do you all think?

HUMANE & SUSTAINABLE FOOD LAB
AT STANFORD UNIVERSITY

Conclusion: meaningfully reducing consumption of meat and animal products is an unsolved problem

- A lot of advocates want a villain, e.g. big bad meat corporations, so we can say, if not for Tyson, we'd be winning, grr!!
- I don't think that's the right model. There's no Lex Luthor here. People just really like meat
- And they don't care enough about how it's raised to pay attention to it or pay higher prices for the better version
- Our challenge is to persuade them otherwise. To quote Lester Freeman on *The Wire*, "Detective, this right here, *this* is the job." It's just going to be a lot of work.

HUMANE & SUSTAINABLE FOOD LAB
AT STANFORD UNIVERSITY